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Strong-Lensing Time Delays in the Berger–Hopf Framework: An R1 Calculation and an Observed-Light Test

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Abstract

Prepared environmental light probes an envelope's state-dependent optical transfer in the Berger–Hopf interpretation. We construct a strong-lensing benchmark from the specified R1 cubic scalar–tensor action and test what ordinary lens freedoms allow its observables to identify. A common source, incident marker, observer clock and distance maps fix the synthetic comparison. A regular finite-body source determines the response without a fitted local amplitude. The declared $10^{11} M_{\odot}$, 3 kpc body has a screening radius of 212.0292 kpc and fixed-body-reference delay offsets of 25.9593 s and 8.6490 s. A positive conventional effective mass reproduces the retained response; a finite conventional refit also matches the injected images and delays. Profile, coefficient and distributed-ray controls quantify the dependencies. We then fit a restricted HE 0435-1223 baseline to 902 historical epochs and reserve 100 later epochs. With later image B as an allowed driver, conventional and conditional R1 predictions for 300 withheld magnitudes differ by only 0.00498 in joint log score. Their nominal 95% intervals cover 85.33% of targets, exposing shared brightness-model limitations. The resulting conditional transfer benchmark and retrospective test establish neither a unique microscopic envelope state nor a BHSM detection.

Keywords: Berger–Hopf framework; strong gravitational lensing; time delays; envelope response; Vainshtein screening; effective-mass representation

1. Introduction

In the Berger–Hopf Standard Model (BHSM) interpretation, gravitational lensing is the optical response of an envelope event traversed by environmental radiation. The BHSM program asks how envelope-state and surface contributions determine that response. Prepared incoming light is related to outgoing image directions and observer arrival-time differences, conditional on the envelope state and surrounding geometry. This raises a modeling question: which properties of a state-conditioned transfer can be identified once ordinary lens and source freedoms are admitted?

We address that question through one explicit conditional realization. The previously specified closed reference branch R1 supplies a cubic scalar–tensor action and frozen rolling coefficients [1]. Its regular finite-body source determines the metric and projected encounter without a fitted local amplitude. The same emitted marker, physical source coordinate, measured redshifts, observer clock and compatible distance maps [2] are used for the response and its conventional fixed-body reference. This common preparation makes their difference a defined forward calculation.

Cubic kinetic braiding, screening and their effects on lensing are established mechanisms [3–5]. Lensing inference also depends on the assumed profile, mass sheet, source

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and propagation geometry [6–9]. Our contribution is a controlled finite-body forward benchmark together with an explicit test of what its image and delay response identifies. It combines the normalized scalar source, one consistent potential/gradient/Hessian, resolved image branches, a positive conventional effective-mass representation and a finite nonlinear nuisance refit. Smooth-profile and distributed-encounter controls test whether the conclusions depend on the convenient body or optical plane. This is the practical role of a bounded forward model with stated limitations [10].

The calculation follows state and preparation, normalized source and boundary problem, metric/optical transfer, images and delays, then identifiability and approximation controls. A positive effective mass reproduces the retained scalar metric and its complete optical response. The reference-relative seconds therefore quantify a conditional transfer; they do not by themselves identify a unique underlying state. The finite refit makes this distinction concrete without changing the injected preparation. We then construct an observed-lens timing and point-geometry baseline for HE 0435−1223 and test frozen conventional and conditional R1 predictions against reserved later light curves. This applies the same identifiability question to actual observations without promoting the synthetic profile to a complete galaxy reconstruction.

BHSM determines the research question and interpretation. The R1 action is an effective representation with its own independent rolling scalar; a unique physical reduction of BHSM cap/interface data to that action requires additional constitutive and embedding data. Those source conditions are derived separately in Supplementary Note S4. The present paper establishes the forward response and its information limits at the level actually specified. Its profiles are controlled examples; the observed-lens application is a restricted retrospective test and does not claim an empirical BHSM detection.

2. Envelope State and Environmental Preparation

The incoming marker probes the event; it does not select the envelope reference state. The conditional state consists of the frozen R1 branch, the finite local density excess and the regular local boundary branch specified below. The earlier source-free target state is retained as a separate null control in Appendix A. Homogeneous geometry, local matter and the local scalar response are accounted for once. This section specifies the synthetic preparation. The distinct observed R-band channel and its measured redshifts are introduced in Section 8; its two hypotheses share that same observational preparation and variability likelihood.

2.1. Incident light, finite body and environment

The lens and source redshifts are $z_l = 0.3$ and $z_s = 1$. The incoming test light is a narrow transient marker with identical emission phase, spectrum, polarization and physical transverse source coordinate in the effective response and its reference. The observed reference wavelength is 550 nm. The fixed source displacement is $s = (0.0014576633674174603, 0)$ Mpc. The light intensity is taken in the test-field limit, so its gravitational backreaction is neglected. The observables are image directions and observer proper-time differences between copies of that marker.

The local body in the conditional envelope is prescribed as a finite excess density above the homogeneous environment. The effective response and fixed-body reference use this same profile:

$$\rho(r) = \frac{15M}{8\pi R^3}\left(1 - \frac{r^2}{R^2}\right) \quad (0 \le r \le R), \qquad M = 10^{11} M_\odot, \quad R = 3 \text{ kpc}. \tag{1}$$

It vanishes outside the body. This is the quadratic-density Tolman–VII profile [11], used as a controlled spherical mass distribution. Define

$$m_g = \frac{GM}{c^2}, \qquad f_M(r) = \frac{M(<r)}{M} = \begin{cases} \frac{1}{2}(5x^3 - 3x^5), & x = r/R \le 1, \\ 1, & r \ge R. \end{cases} \tag{2}$$

For the effective scalar response, the density is prescribed and the leading pressure adjustment is calculated separately. We do not reuse the exact Einstein-fluid pressure as an exact modified-gravity solution.

The physical Jacobi maps are independently initialized for observer–lens, observer–source and lens–source propagation. They share one background, the same transported screen basis, and frequency normalization. At the reference state their scalar values are

$$(D_{ol}, D_{os}, D_{ls}) = (945.162839373,\ 1689.312239687,\ 1076.994348060)\ \text{Mpc}. \tag{3}$$

The main body is excluded from these maps and enters the local encounter once. The local response calculation holds the maps fixed, isolating the encounter contribution. Its fixed-body reference uses the same R1 geometry; it is not a fit of a separate complete ΛCDM cosmology.

3. Action, Normalization, Local Source and Boundary Data

3.1. Conditional BHSM state representation

The question is how an envelope state determines a light transfer, and which state information that transfer identifies. At the effective level used here, the state specifies the rolling values H, v, d, the cubic normalization λ, the finite matter source and the regular boundary branch. The normalized action below makes their source and metric applications explicit. This is the state-conditioned forward problem used throughout the paper.

A unique microscopic map from BHSM cap/interface variables into this independent scalar action is not assumed. The cap momentum and source pullback, including their normalization and complement dependence, are worked out in Supplementary Note S4. They specify additional physical selection data rather than another adjustable lensing contribution. The present light calculation measures properties of the conditional metric response. Its equivalence to a conventional effective mass is then a statement about what those observations can identify.

3.2. Effective action and frozen R1 normalization

We use signature $(-,+,+,+)$ and $M_{\rm Pl}^{-2} = 8\pi G$. In the action and local field equations, $c = \hbar = 1$. The conditional R1 representation uses the retained scalar–tensor action

$$S = \int \sqrt{-g}\, \mathrm{d}^4x \left[\frac{M_{\rm Pl}^2}{2} R - \frac{1}{2}(\nabla T)^2 + \frac{(\nabla T)^2 \Box T}{\lambda} - V_0 + \mu^3 T + \mathcal{L}_m \right], \qquad \lambda = \Lambda^3 > 0. \tag{4}$$

The field T is an independent scalar. It is distinct from the target field η and from a metric conformal coordinate. For example, $\gamma_{ij} = a^2 e^{2\psi_{\rm geo}} \bar{\gamma}_{ij}$ gives $\psi_{\rm geo} = -\Psi$ to linear order in Newtonian gauge; it does not supply another independent scalar in addition to the metric.

The dimensionless background convention is $\widehat{T} = T/M_{\rm Pl}$, $\widehat{v} = \dot{\widehat{T}}/(M_{\rm Pl}H_0)$, $\widehat{d} = \ddot{\widehat{T}}/(M_{\rm Pl}H_0^2)$, $\widehat{\lambda} = \lambda/(M_{\rm Pl}H_0^2)$. The frozen parameters are

$$(\Omega_{m0}, \Omega_{r0}, \Omega_{k0}) = (0.31, 9\times10^{-5}, -0.018),$$
$$H_0 = 67.4\ \mathrm{km\,s^{-1}\,Mpc^{-1}}, \qquad \widehat{\lambda} = 1, \qquad \frac{\mu^3}{M_{\rm Pl}H_0^2} = 3,$$
$$\frac{V_0}{M_{\rm Pl}^2H_0^2} = 2.3986073449708067. \tag{5}$$

They define one reference branch, not a framework-wide calibration. The stored background solution is reused without refitting. R1 serves as a reproducible fiducial background for this transfer benchmark; no fit to current cosmological observations or claim of phenomenological viability is made here. The reference spatial curvature is $K = -\Omega_{k0}H_0^2/c^2 > 0$. Its unperturbed angular-diameter maps obey $D_{ab} = a_b \sin[\sqrt{K}(\chi_b - \chi_a)]/\sqrt{K}$. The previously computed six-amplitude sensitivity of one normalized axial $n = 2$ harmonic is retained as a separate propagation control; its amplitudes are zero in the reference maps used for the local comparison. Its homogeneous scalar current,

$$J = \dot{\bar{T}} + \frac{6H\dot{\bar{T}}^2}{\lambda}, \qquad \dot{J} + 3HJ = \mu^3, \tag{6}$$

also provides a sign and normalization check on the local variation.

3.3. Reduction from the retained effective action

Varying Equation (4) gives

$$\Box T + \frac{2}{\lambda}\Big[(\nabla_\mu\nabla_\nu T)^2 - (\Box T)^2 + R_{\mu\nu}\nabla^\mu T\nabla^\nu T\Big] - V_T = 0. \tag{7}$$

The Ricci term couples the scalar response to matter through the same Einstein equation. No independent phenomenological matter coupling is introduced. Write $T = \bar{T}(t) + \pi_T(r)$ and freeze $v = \dot{\bar{T}}$, $d = \ddot{\bar{T}}$ and H at the lens epoch. After subtracting the homogeneous equations, retain leading weak metric potentials and quasi-static derivatives, including the nonlinear spatial Hessian responsible for screening. Set

$$b_T = \frac{2v^2}{\lambda}, \qquad C = 1 + \frac{4(d+2Hv)}{\lambda} - \frac{2v^4}{M_{\rm Pl}^2\lambda^2}. \tag{8}$$

The leading common constraints are

$$2M_{\rm Pl}^2\Delta\Phi = \delta\rho - b_T\Delta\pi_T, \qquad \Phi = \Psi, \tag{9}$$
$$C\Delta\pi_T - \frac{2}{\lambda}\Big[(\Delta\pi_T)^2 - (\partial_i\partial_j\pi_T)^2\Big] = -\frac{b_T}{2M_{\rm Pl}^2}\delta\rho. \tag{10}$$

The last term of C follows from eliminating the metric constraint. The equal potentials are a result at the retained order; they are not an exact assertion about a fully nonlinear space-time.

At $z_l = 0.3$ the frozen solution yields

$$\begin{aligned} H/H_0 &= 1.1836105772, & \widehat{v} &= 0.2377644264,\\ \widehat{d} &= 0.2484715891, & b_T/M_{\rm Pl} &= 0.1130638449,\\ C &= 4.2388585594. \end{aligned} \tag{11}$$

The physical b_T has mass dimension one. In numerical units $\widehat{\lambda} = 1$ is fixed; it is not a new local adjustable length.

3.4. Regular finite-body branch

For spherical symmetry the scalar equation integrates to

$$Cr^2\pi_T' - \frac{4}{\lambda} r(\pi_T')^2 = -b_T GM(<r). \tag{12}$$

The origin is regular and carries no additional scalar charge. The root continuously connected to the decaying weak-source solution is

$$\pi_T' = -\frac{2b_T GM(<r)}{r^2\left[C + \sqrt{C^2 + 16 b_T GM(<r)/(\lambda r^3)}\right]}. \tag{13}$$

It has $\pi_T' \propto r$ at the center and no interface delta-function source at R.

3.5. Executed source and material-boundary application

The source consumed by this reduction can be made explicit. Define an additional metric energy source ρ_Σ and independent scalar Euler source J_T by the pre-elimination rows $2M_{\rm Pl}^2\Delta\Phi = \rho + \rho_\Sigma - b_T\Delta\pi_T$ and $A\Delta\pi_T - 2[(\Delta\pi_T)^2 - |\,\mathrm{Hess}\,\pi_T|^2]/\lambda + b_T\Delta\Phi = J_T$, where $A = C + b_T^2/(2M_{\rm Pl}^2)$. The actual reduced source and spherical flux are

$$J_{\rm eff} = J_T - \frac{b_T}{2M_{\rm Pl}^2}(\rho + \rho_\Sigma),$$

$$Cr^2\pi_T' - \frac{4}{\lambda} r(\pi_T')^2 = \int_0^r s^2 J_{\rm eff}(s)\,\mathrm{d}s. \tag{14}$$

For the declared action there is no extra layer term: $\rho_\Sigma = J_T = 0$, while the volume source remains nonzero. For a proper-radial thin layer with coefficients σ_Σ and $j_{T,\Sigma}$, its retained weak-metric jump would instead be

$$[Cr^2\pi_T' - 4r(\pi_T')^2/\lambda]_{R^-}^{R^+} = R^2\left(j_{T,\Sigma} - \frac{b_T\sigma_\Sigma}{2M_{\rm Pl}^2}\right). \tag{15}$$

These are source applications of the original coupled equations; a surface charge is not an independently adjustable lensing amplitude. An additional anisotropic interface stress would also consume the corresponding metric slip equation.

Using the frozen coefficients, the computed surface section is $\pi_T'(R)/(M_{\rm Pl}H_0) = -2.119825778845613 \times 10^{-4}$. In $M_{\rm Pl} = H_0 = c = 1$ units, with radius in c/H_0, an independent density integration gives $\int_0^R r^2 J_{\rm eff}\,\mathrm{d}r = -1.216417408185084 \times 10^{-13}$. The maximum relative nonlinear-flux residual at $r/R = 0.1, 0.5, 1, 2, 10$ is 4.15×10^{-16}; the original integrated metric Gauss residual is zero at printed precision. Independent interior quadratic and exterior rational roots give a relative material-boundary flux mismatch 6.23×10^{-16}. The separate exact-GR material control has continuous same-oriented extrinsic curvature and no material layer. That material worldtube is distinct from the cosmological cap seam, and its zero layer does not remove the nonzero volume source. The receipt is `artifacts/strong_lensing_bhsm_integration_20261020/source_application.json`. Supplementary Note S4 derives the cap/source pullback and shows constructively why the leading normalization does not select a unique microscopic completion.

4. Metric and Optical Encounter Response

The common metric constraint gives

$$\Phi' = \Psi' = \frac{GM(<r)}{r^2} - \frac{b_T}{2M_{\rm Pl}^2}\pi_T', \qquad F(r) \equiv \frac{\delta\Phi'}{\Phi_N'} = \frac{2\epsilon_\infty}{1+\sqrt{1+r_V^3 f_M(r)/r^3}}, \tag{16}$$

where

$$\epsilon_\infty = \frac{b_T^2}{2M_{\rm Pl}^2 C}, \qquad r_V^3 = \frac{16 b_T GM}{\lambda C^2}. \tag{17}$$

The last expression uses $c = 1$. The numerical conversion is

$$r_V^3 = 16(b_T/M_{\rm Pl})m_g(c/H_0)^2/(\widehat{\lambda}C^2).$$

Deep in the exterior screened region $F(r)$ approaches $2\epsilon_\infty(r/r_V)^{3/2}$; at large radius it approaches ϵ_∞. The potential itself still decays. This is the usual cubic screening mechanism [4,5], evaluated here with the frozen normalization and a regular extended body.

The corresponding leading hydrostatic correction is

$$\delta p(r) = \int_r^R \rho(s)\,\delta\Phi'(s)\,{\rm d}s, \qquad \delta p(R) = 0. \tag{18}$$

It is positive and changes the central Newtonian pressure by 3.56959×10^{-6} fractionally. This supports the specified leading-order matter configuration; it does not construct an exact modified Tolman–Oppenheimer–Volkoff sequence or prove radial stability. Holding density fixed and changing its required pressure does not impose an independently fixed common material equation of state on both models.

4.1. Projection into the optical encounter

Restore c and interpret Φ as the dimensionless metric potential. For physical impact radius $b_{\rm imp}$ the additional deflection is

$$\delta\hat{\alpha}(b_{\rm imp}) = 4b_{\rm imp}\int_0^\infty \frac{\delta\Phi'(\sqrt{b_{\rm imp}^2+z^2})}{\sqrt{b_{\rm imp}^2+z^2}}{\rm d}z, \tag{19}$$

$$\delta\hat{\psi}(b_{\rm imp}) = \int_R^{b_{\rm imp}} \delta\hat{\alpha}(s)\,{\rm d}s. \tag{20}$$

The arbitrary zero of potential does not affect relative delays. We calculate the gradient and Hessian consistently from this same projected potential. A local force ratio is not substituted for a projected photon deflection ratio.

The leading stationary delay change between image i and reference image 0 is

$$\delta\Delta t_{i0} = -\frac{1+z_l}{c}\left[\delta\hat{\psi}(b_i) - \delta\hat{\psi}(b_0)\right]. \tag{21}$$

The finite result also resolves the image positions in Equation (23). The projection freezes the local coefficients during the encounter and extends the isolated local solution through the thin plane. Matching to the linear perturbation is assumed in an overlap region $r_V \ll r \ll c/H_l$; a global closed-cosmology solution has not been constructed.

4.2. Which state coefficients enter the transfer?

The state dependence is explicit in $b_T = 2v^2/\lambda$ and $C = 1 + 4(d + 2Hv)/\lambda - b_T^2/(2M_{\rm Pl}^2)$. At fixed body, λ and physical radius, define $S_r = \sqrt{1 + r_V^3 f_M(r)/r^3}$. Differentiating the original force gives

$$\frac{\partial \ln \delta\Phi'}{\partial \ln b_T} = \frac{3}{2} + \frac{1}{2S_r}, \qquad \frac{\partial \ln \delta\Phi'}{\partial \ln C} = -\frac{1}{S_r}. \tag{22}$$

Deep screening suppresses the local force sensitivity to C. Projection in Equation (19) takes positive weighted averages of these coefficients. For the reference image impacts, ordered positive, central and negative, the projected logarithmic derivatives are

$$\begin{aligned} \partial_{\ln b_T} \ln \delta\hat{\alpha} &= (1.525461, 1.516634, 1.522208), \\ \partial_{\ln C} \ln \delta\hat{\alpha} &= (-0.050923, -0.033268, -0.044417). \end{aligned}$$

The outer field therefore retains nonzero C sensitivity even though the impact lies well inside r_V.

These derivatives concern the extra deflection; derivatives divided by the total deflection carry its additional small fractional contribution. They quantify the conditional state-to-transfer relation without changing the frozen prediction. A particular physical state variation must also satisfy the action's background and boundary constraints and be composed with its H, v, d to b_T, C map. Independently varying the two reduced coefficients is not asserted to produce another globally matched R1 state. Section 6 addresses the separate question of identifying those coefficients from light when the ordinary mass distribution is allowed to change.

5. Image Directions and Observer Delays

5.1. Image directions and observer arrival-time differences

Let $x = D_{ol}\theta$, $\beta = D_{os}^{-1}s$ and $\hat{\alpha} = \nabla_x \hat{\psi}$, where $\hat{\psi}$ has length units. The common thin-lens equations are

$$\beta = \theta - D_{os}^{-1} D_{ls} \hat{\alpha}(D_{ol}\theta), \tag{23}$$

$$ct(\theta) = \frac{1}{2}(\theta - \beta)^T \mathsf{T} (\theta - \beta) - (1 + z_l)\hat{\psi}(D_{ol}\theta) + f(\beta),$$

$$\mathsf{T} = (1 + z_l) D_{ol}^T D_{ls}^{-1} D_{os}. \tag{24}$$

The maps must satisfy reciprocity so that T is symmetric [2]. A common $f(\beta)$ cancels in relative delays. Image positions are resolved again after changing the encounter.

For smooth geometric variations, fixing s implies $\delta\beta = -D_{os}^{-1}\delta D_{os}\beta$; fixed physical and fixed angular source prescriptions cannot be imposed simultaneously. At fixed measured redshift, the stationary first variation is

$$c\delta t = \frac{1}{2} q^T \delta\mathsf{T} q - q^T \mathsf{T} \delta\beta - (1 + z_l)\left[\hat{\alpha}^T \delta D_{ol}\theta + \delta\hat{\psi}|_x\right] + \delta f, \qquad q = \theta - \beta. \tag{25}$$

The image-position variation cancels at a stationary image. In the local R1 comparison $\delta D_{ab} = \delta\beta = 0$ and the nonzero intrinsic term is retained. Appendix B records the measured-redshift convention for comparisons that vary the maps.

5.2. Evaluated response and reference-relative offsets

The screened branch predicts $r_V = 212.0292$ kpc and $\epsilon_\infty = 0.001507886$, or a 0.1507886% asymptotic force enhancement. At $r = R = 3$ kpc the local force enhancement

is 5.06706 parts per million (ppm), whereas the deflection enhancement at the same impact radius is 11.7467 ppm. The ray samples the outer, less screened field as well as the field near closest approach.

Table 1. Leading-order image positions for the effective response and its fixed-body reference. Image 0 is the positive outer image, image 1 the central image, and image 2 the negative outer image. Printed digits specify reproducible model values, not absolute astrometric accuracy.

Image	Reference (arcsec)	R1 response (arcsec)	Offset (μas)
0	$+0.835476034$	$+0.835482157$	$+6.1233$
1	-0.082161940	-0.082161152	$+0.7881$
2	-0.657495525	-0.657499874	-4.3486

Table 2. Relative delays for the same source marker, with image 0 as reference. The offsets are relative to the specified fixed-body reference, with the physical light preparation held fixed.

Observable	Fixed-body reference	R1 response	Offset
Δt_{10} (days)	21.716076566	21.716377020	25.9593 s
Δt_{20} (days)	14.271452387	14.271552491	8.6490 s
$\Delta t_{10}/\Delta t_{20}$	1.521644467	1.521654847	6.8212 ppm

Figure 1 compares the radial response and the two delay changes. The nonzero response follows from the rolling scalar already in Equation (4). It does not alter the earlier constant-target null result in Appendix A. An independent singular isothermal control, with velocity dispersion 220 km s^{-1} and the same reference source, retains images at $+1.067883$ and -0.711922 arcsec and a 17.094532-day delay. That analytic control validates the optical conventions; the finite body in Tables 1 and 2 is the local boundary problem used for the evaluated response.

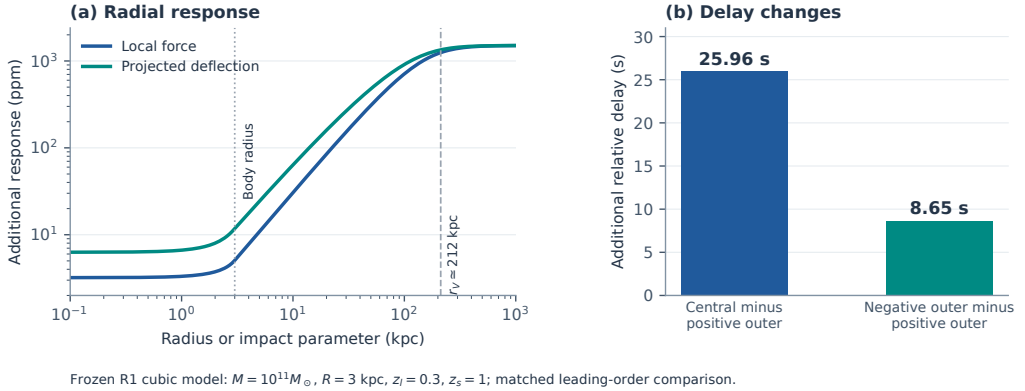


Figure 1. R1 envelope response relative to the fixed-body reference. Left: additional local force and projected deflection as fractions of their fixed-body reference values. The distinction reflects integration along the ray. Right: reference-relative offsets in the two delays for the fixed source preparation. Neither panel is an observational detection forecast.

At fixed epoch, $r_V \propto M^{1/3}$. The same frozen background also specifies the redshift dependence, without refitting:

Lens redshift	ϵ_∞	r_V (kpc), $M = 10^{11} M_\odot$
0.0	0.00351132	248.300
0.3	0.00150789	212.029
0.5	0.00086052	191.230
1.0	0.00022832	150.778
1.5	0.00007085	123.011

This is a prediction for the coefficients and force law, not a delay table for identical angular sources at these different redshifts.

The supplied reference-run outputs give the following implementation checks. The integrated scalar flux has maximum relative residual 7.9×10^{-16}. A direct line-of-sight quadrature and an independent angular-substitution quadrature agree on the additional deflection to 4.5×10^{-13} relative. The image equation residual is below 8.5×10^{-22} rad. Differentiating the projected potential and deflection independently gives relative discrepancies below 2.0×10^{-9} and 2.4×10^{-12}, respectively. Finite re-solved and stationary first-order delay changes differ by 0.00016 s and 0.000056 s. These are numerical and perturbative checks within the model, not bounds on omitted physics. In the current Windows replay, the corresponding flux residual is 8.30×10^{-16} and the deflection-derivative discrepancy is 3.19×10^{-12}; the remaining differences are recorded with their numerical tolerances in the replay receipt. The supplied and replay outputs remain separate records.

6. What the Light Transfer Identifies

6.1. Positive effective-mass representation of the same response

The retained metric and optical response admit an explicit conventional representation. Define an effective additional enclosed mass by

$$\delta M_{\rm eff}(r) = M f_M(r) F(r), \qquad \delta\rho_{\rm eff}(r) = \frac{M}{4\pi r^2}\frac{\mathrm{d}}{\mathrm{d}r}[f_M(r)F(r)]. \tag{26}$$

This density is nonnegative. To see this, put $S = \sqrt{1 + r_V^3 f_M / r^3}$ and $n_M = \mathrm{d}\ln f_M / \mathrm{d}\ln r \geq 0$. Then

$$\frac{\mathrm{d}\ln(f_M F)}{\mathrm{d}\ln r} = \frac{1}{2}\Big[(1+S^{-1})n_M + 3(1-S^{-1})\Big] \geq 0. \tag{27}$$

The ordinary Poisson field of $M(<r) + \delta M_{\rm eff}(r)$ has exactly the same $\Phi = \Psi$ as Equation (16) within the retained approximation. It therefore reproduces the full deflection field, relative potential, images and delays, including multiple source planes in the same geometry. Its asymptotic extra mass is

$$\delta M_{\rm eff}(\infty) = \epsilon_\infty M = 1.50788624 \times 10^8 M_\odot. \tag{28}$$

The rank-one multiple-source residual discussed below distinguishes the scalar from the restricted Tolman–VII parameter family, not from all conventional matter distributions. Because slow matter and light respond to the same retained potentials, ordinary dynamical data do not automatically break this equivalence either. Independent restrictions on the matter/environment model or an observable outside this retained equivalence would be needed to distinguish those representations. No such independent likelihood is asserted here, and such a distinction is not required for the envelope-transfer calculation.

The effective mass in Equation (26) represents the same retained contribution as the scalar correction in Equation (16). One uses either the explicit scalar correction or its effective-mass representation in the encounter, not their sum. The construction proves equivalence of the retained metric and optical response. The physical pullback of micro-

scopic BHSM surface variables into the normalized effective source is specified separately in Supplementary Note S4.

6.2. Single-source constraints and nuisance freedoms

For an observable residual $\delta y = y - y_0$, write

$$\delta y = B\,\delta a + N\,\delta \nu + \epsilon, \qquad WCW^T = I, \qquad P = I - (WN)(WN)^{+}. \tag{29}$$

Here B is a physical response, N the stated nuisance response, C a specified covariance, and $+$ a Moore–Penrose inverse. The rank of PWB concerns distinguishability in that nuisance model. A zero direction is unconstrained by the corresponding likelihood; a Fisher pseudoinverse does not turn it into a zero error bar. We take C positive definite. For the screened comparison, the single column is a fixed reference-relative response template; rank one denotes a surviving residual direction, not identification of a freely varied fundamental scalar parameter.

Using source position, body mass and radius, and a compensated mass-sheet scale as nuisances, the single R1-response column has projected rank one if the body center is fixed. Adding both center coordinates gives rank zero. The raw delay shifts in Table 2 are therefore not nuisance-independent observables for this configuration. The rank calculation uses 0.001 arcsec and 0.1 day as numerical whitening scales, not as demonstrated observational precisions.

The ordinary mass-sheet transformation,

$$\psi_\lambda(\theta) = \lambda\psi(\theta) + \frac{1-\lambda}{2}|\theta|^2, \qquad \beta_\lambda = \lambda\beta, \qquad \Delta t_\lambda = \lambda\Delta t, \tag{30}$$

cancels from the ratio of two delays. It does not remove source, centering, radial-profile, or independent line-of-sight freedoms [8]. Consequently the 6.8212 ppm ratio change is a fixed-preparation prediction, not a unique diagnostic of a distinct representation.

For the separately computed axial smooth-map response, the images of a softened isothermal model depend only on $\beta = s/D_{os}$, $\theta_E = b_{\rm phys} D_{ls}/D_{os}$, and $\theta_c = r_c/D_{ol}$, while all delays share the factor $(1+z_l)D_{ol}D_{os}/(cD_{ls})$. Varying the three scalar maps can therefore be reproduced by these source/profile changes and a delay scale. The saved six-anchor response accordingly has zero projected rank in the unrestricted nuisance model. This analytic factorization concerns that family; it is not a universal theorem that cosmological lensing information always vanishes.

6.3. A finite nonlinear conventional refit

A local rank calculation can be tested by a finite inverse application. We take the saved R1 images and delays as a deterministic injection and fit an ordinary Tolman–VII body with mass, radius, source position, lens center and one physical convergence sheet free. The injected marker, state, geometry and source remain the original preparation; fitted source coordinates are inverse-model nuisances. The fitted encounter contains the sheet once, while the propagation maps remain fixed. No independent delay scale is introduced.

For each of the three observed axial image positions, the original rows are $\theta_i - \beta - q\hat{\alpha}_{M,R}[D_{ol}(\theta_i - \theta_c)] - \kappa(\theta_i - \theta_c) = 0$, with $q = D_{ls}/D_{os}$. The two relative Fermat arrival rows use

$$\hat{\psi}(x) = \hat{\psi}_{M,R}(x - D_{ol}\theta_c) + \frac{\kappa}{2D_{ol}q}|x - D_{ol}\theta_c|^2. \tag{31}$$

All three image branches are re-solved after fitting. This is a bounded five-coordinate fit to five axial observables, a deliberately saturated local example whose Jacobian is nonsingular. It demonstrates finite absorption in this example; it is not a global uniqueness or detection

claim. The free-center original lens and stationarity residuals are 4.24×10^{-22} rad and

Table 3. Finite conventional fits to the saved R1 injection. Entries are numerical mismatches after re-solving image branches, not observational errors or sensitivity forecasts.

Quantity	Center fixed	Center free
$\delta \ln M$	3.9960×10^{-6}	4.0206×10^{-6}
$\delta \ln R$	3.9827×10^{-7}	4.1234×10^{-7}
$\delta\beta$ (arcsec)	4.3495×10^{-9}	6.8148×10^{-9}
θ_c (arcsec)	0	1.7743×10^{-9}
κ	9.9469×10^{-6}	9.9229×10^{-6}
Maximum image mismatch (μas)	1.0143×10^{-4}	8.74×10^{-11}
First delay mismatch (s)	$+0.0033064$	$+9.21 \times 10^{-10}$
Second delay mismatch (s)	-0.0050312	-1.53×10^{-10}

1.30×10^{-18} Mpc. Allowing the ordinary model to adjust therefore removes the original 25.96 s and 8.65 s fixed-body offsets to numerical precision in this preparation. Fixing the center leaves a small finite residual; its significance would require an actual observing likelihood and independently justified lens priors. Supplementary Note S7 gives the original equations, parameter bounds, mass-sheet symmetry and independent-root checks.

6.4. A bounded multiple-source test

A supplemental design uses one body and three sources at $z_s = (0.8, 1.2, 1.8)$, with source offsets $0.1\theta_{E,s}$ at angles $(0.15, 1.1, 2.2)$ rad. These choices were fixed before applying the screened response. The nuisance model includes independent source positions and delay scales, a shared mass, radius and center, and shared physical convergence and two shear components. The resulting scalar column retains rank one; adding independent source-plane convergence columns leaves rank one as well.

This surviving component is small: its norm is 0.1202% of the raw whitened response in the expanded nuisance model. In the same illustrative weighting, the residual coordinates reach approximately 6 nanoarcsec and the residual delays 0.086 s. The explicit 24 observable weights and response arrays are supplied with the code. These tidal nuisance fields are phenomenological controls. This test does not solve the nonlinear scalar equation for a nonspherical galaxy and its environment, and it is not a signal-to-noise forecast. Its rank is specific to the finite radial profile family.

7. Robustness of the Transfer Calculation

7.1. A smooth profile with the same mass and half-mass radius

The body is a controlled input, so the response must be tested against its profile choice. We compare the compact Tolman–VII body with the standard smooth Plummer sphere [12], matching total three-dimensional mass and three-dimensional half-mass radius, rather than fitting a desired deflection. The reference half-mass radius is $r_h = 1.92941634649$ kpc. The Plummer scale is fixed by $a = r_h\sqrt{2^{2/3} - 1} = 1.47874508325$ kpc, with

$$\rho_P(r) = \frac{3Ma^2}{4\pi(r^2 + a^2)^{5/2}},$$
$$M_P(< r) = \frac{Mr^3}{(r^2 + a^2)^{3/2}}, \qquad \hat{\alpha}_P(b) = \frac{4m_g b}{b^2 + a^2}. \tag{32}$$

This profile has a regular center and finite total mass, with no hard material edge. Its consistent potential is $\hat{\psi}_P = 2m_g \ln[(b^2 + a^2)/(2a^2)]$. The same original scalar flux, regular root and metric constraint are applied to its enclosed mass. The background, clock and

maps remain fixed. Both profiles have the same reference r_V because their total mass and R1 state are equal.

Eight paired applications use the two profiles and four settings: reference, total mass increased by 20%, half-mass radius increased by 20%, or physical source displacement increased by 20%. Only one ordinary input changes at a time, and the change is shared by the response and its same-profile reference. These are sensitivity controls, with no assigned probability or empirical uncertainty prior. All eight applications retain three images.

Table 4. Re-solved reference-relative delay offsets for two declared profiles. Each pair uses common preparation, state and ordinary inputs. The finite input changes are controls, not fitted uncertainties.

Shared control	Tolman–VII offsets (s)	Plummer offsets (s)
Reference	25.9593, 8.6490	22.4510, 9.6200
M increased 20%	32.6999, 9.8907	28.6743, 10.7843
r_h increased 20%	24.2697, 9.9458	20.2722, 10.1974
Source offset increased 20%	26.8452, 10.3945	23.5110, 11.5634

At matched M, r_h, changing the ordinary profile changes the two reference delays by approximately -167125 s and -124500 s, while changing the R1 offsets by -3.5083 s and $+0.9710$ s. The former are deliberate model changes, about 6438 times the largest reference R1 offset; they are not an estimated error on an observed lens. This shows why mass and one size alone do not fix the transfer. A ray samples the entire field, even though both profiles share the same scalar force at their half-mass radius.

The positive effective-mass construction applies to both profiles. Independent direct line-of-sight projection and re-solved images agree with their scalar representations within 5.56×10^{-17} arcsec and 6.14×10^{-10} s. The maximum original scalar-flux residual across the eight applications is 7.25×10^{-16} and the maximum image-equation residual is 2.55×10^{-21} rad. Both baseline profiles retain projected rank one with a fixed center and zero with a free center in the declared linearized nuisance family. Thus profile smoothness changes the quantitative response while preserving its retained conventional representation. Supplementary Note S6 gives the profile normalization, support and independent checks.

7.2. The ordinary finite-body control

For the conventional control, a regular Einstein-fluid interior can be matched to Schwarzschild. Its compactness is $m_g/R = 1.59514 \times 10^{-6}$. The leading optical application above uses the weak-field projection of this density.

7.3. A common second-order ordinary-gravity control

To test the effect of the first omitted conventional deflection, we expand the exact static GR optical scattering integral of the same Tolman–VII body through second post-Minkowskian order (2PM). Put $\varepsilon_g = m_g/R$, $y = b_{\rm imp}/R$ and $t = \sqrt{1-y^2}$. The additional radial deflection is

$$\delta\hat{\alpha}_{\rm GR2}(b_{\rm imp}) = 2\varepsilon_g^2 I(y), \tag{33}$$

$$I(y) = \begin{cases} \dfrac{15(\arcsin y - yt)}{4y^2} + y\left(\dfrac{15}{2}t - 4t^5 - \dfrac{48}{7}t^7\right), & 0 < y < 1, \\ \dfrac{15\pi}{8y^2}, & y \geq 1. \end{cases} \tag{34}$$

At the center $I(0) = 0$ and $I'(0) = -6/7$; the interior coefficient includes the leading fluid pressure and the conversion from areal radius to invariant asymptotic impact. At and

outside the surface it matches the standard $15\pi m_g^2/(4b_{\rm imp}^2)$ exterior deflection [13]. The consistent plane potential is

$$\delta\hat{\psi}_{\rm GR2}(b_{\rm imp}) = 2\varepsilon_g^2 R J(y), \qquad J'(y) = I(y), \qquad J(1) = 0. \tag{35}$$

The supplied implementation evaluates I, I' and J analytically.

The R1 response and fixed-body reference receive the identical GR2 potential, gradient and Hessian, and both image systems are re-solved. The R1 coefficients and leading scalar profile are held fixed. The scattering impact is identified with the physical thin-plane impact in this control. This checks robustness to the common GR term; it does not compute the missing scalar–GR higher-order terms or a complete finite-distance 2PM cosmological arrival functional. The replay command is

```
python scripts/run_strong_lensing_gr2_control.py
```

7.4. Result of the common GR correction

At the original two exterior-image impacts, the GR2 term is 3.68 and 4.68 ppm of the leading conventional deflection. Its inclusion changes the conventional image coordinates by $(+1.35456, -0.01519, -1.72123)$ microarcsec and the conventional relative delays by $(+8.28473, +2.55245)$ s. The absolute correction is therefore material at the scale of the calculated reference-relative image shifts.

The response-minus-fixed-body-reference offsets are considerably more stable under this common-term check.

Table 5. Robustness of the reference-relative R1 offsets under the same GR2 correction in both calculations. The final column is the change in the reference-relative offset, not the absolute GR correction.

Response offset	Leading calculation	Common GR2 added	Change
$\delta\Delta t_{10}$ (s)	25.95925815	25.95932714	$+68.99$ μs
$\delta\Delta t_{20}$ (s)	8.64901938	8.64900924	-10.14 μs
$\delta\theta_0$ (μas)	$+6.12327625$	$+6.12326934$	-6.91×10^{-6} μas
$\delta\theta_1$ (μas)	$+0.78810716$	$+0.78810711$	-5.58×10^{-8} μas
$\delta\theta_2$ (μas)	-4.34859153	-4.34858643	$+5.09 \times 10^{-6}$ μas

The two relative changes in the reference-relative delay offsets are $+2.66$ and -1.17 ppm. The delay-ratio response remains 6.8212 ppm. Thus the nonzero reference-relative response survives this specific common-term control without changing its quoted precision in seconds or microarcseconds.

An independent mixed stationary-action derivative reproduces the change in the reference-relative delay offsets within 0.7 ns; this checks the small drift without subtracting four re-solved delays.

The analytic correction was also compared with direct null-geodesic scattering in the solved static GR fluid and exterior metric. At the three reference impacts, the exact deflection minus the 1PM and 2PM terms has magnitude below 1.8×10^{-16} rad for this body's compactness $\varepsilon_g = 1.59514 \times 10^{-6}$. A compactness extrapolation independently recovers the interior 2PM coefficient, and direct integration verifies $J' = I$. The latter checks establish the implemented ordinary-GR scattering coefficient, not a physical error bound for the scalar model or its cosmological embedding. The full corrected coordinates, delays and checks are supplied in `gr2_control.json` and the review outputs.

7.5. A distributed encounter with the same physical endpoints

The thin plane can be checked by distributing the same retained field along the ray. Use lens-scaled conformal coordinates $z = a_l(\chi - \chi_l)$, $x = a_l x_{\rm com}$ and $\kappa_K = K/a_l^2$, where $a_l = (1+z_l)^{-1}$. The observer and source lengths are $\ell_o = a_l\chi_l$ and $\ell_s = a_l(\chi_s - \chi_l)$; the fixed source endpoint is $X_s = (a_l/a_s)s$. These are determined by the original closed reference and its physical distance maps. Define $S_\kappa(d) = \sin(\sqrt{\kappa_K}d)/\sqrt{\kappa_K}$, $C_\kappa(d) = \cos(\sqrt{\kappa_K}d)$ and the background generating function

$$G_d(x_1, x_2) = \frac{C_\kappa(d)(x_1^2 + x_2^2) - 2x_1x_2}{2S_\kappa(d)}. \tag{36}$$

For a common half-window L, the retained paraxial arrival action is

$$ct[x] = \frac{1}{a_l}\Bigg\{ G_{\ell_o - L}[0, x(-L)] + G_{\ell_s - L}[x(L), X_s] + \int_{-L}^{L}\left[\frac{x'^2 - \kappa_K x^2}{2} - 2\Phi(\sqrt{x^2+z^2})\right]\mathrm{d}z\Bigg\}. \tag{37}$$

Its original ray equation and endpoint rows are

$$\begin{aligned} x'' + \kappa_K x &= -2\partial_x\Phi, \\ x'(-L) &= \frac{C_\kappa(\ell_o - L)}{S_\kappa(\ell_o - L)}x(-L), \\ x'(L) &= -\frac{C_\kappa(\ell_s - L)}{S_\kappa(\ell_s - L)}x(L) + \frac{X_s}{S_\kappa(\ell_s - L)}. \end{aligned} \tag{38}$$

The individual exterior wavefront curvatures are fixed by those physical endpoint data. At the plane their sum is $D_{os}/(D_{ol}D_{ls})$ and the source term is s/D_{ls}, so they recover the original thin-plane preparation. They are not a freely chosen split of that sum. The factor $1/a_l = 1 + z_l$ retains the same observer clock.

Both the fixed-body and R1 fields are solved on each window, using all three noncritical ray branches. Their distributed actions are compared with the thin-plane/Born projection of the same windowed fields. At $L = 20$ Mpc, the same-window thin-plane offsets are

Table 6. Distributed-path change of the reference-relative delay offsets compared with the same-window thin plane, evaluated by direct arrival subtraction. This is a control of the retained frozen weak-field optical action.

Half-window L (Mpc)	Change of $\delta\Delta t_{10}$ (ms)	Change of $\delta\Delta t_{20}$ (μs)
0.21203	−0.46019	+1.5548
0.84812	−0.57857	+1.1474
4.24058	−0.61238	+1.0321
20.00000	−0.61905	+1.0056

25.95922277 s and 8.64900576 s; the distributed offsets are 25.95860372 s and 8.64900676 s. Tighter collocation and potential quadrature change the latter offsets by less than 0.3 ns. These are refinement figures for that subtraction. A separate on-shell derivative integrated along a numerical interpolation between the two fixed optical fields gives the preferred finite-response changes −0.61904348 ms and +1.00640532 microseconds. They agree with the raw subtraction within 7 ns; the interpolation is a numerical integration device, not a physical state variation. These are numerical controls within this defined optical problem.

An analytic exterior-tail bound follows from $0 \le \delta\Phi' \le \epsilon_\infty m_g/r^2$. For two fixed Born impacts b_i, b_0, it gives

$$|\delta\Delta t_{i0}^{\rm tail}| \le \frac{1+z_l}{c}\,\frac{m_g\epsilon_\infty|b_i^2-b_0^2|}{L^2}. \tag{39}$$

The sharper integrated bounds at 20 Mpc are 35.04 and 13.47 microseconds for the two extra-field fixed-impact delays. They do not bound shifts caused by re-solving all images, an arbitrary environment, or a changed global state. The new control isolates finite thickness and path response with fixed background, source and clock. It does not add the scalar-dependent field-equation corrections needed for a full higher-order metric prediction. The complete boundary construction, force/action residuals, tail derivation and convergence data are in Supplementary Note S8.

8. An Observed-Lens Baseline and Withheld-Light Test

8.1. Data, calibration and reserved observations

We next apply the transfer-to-identifiability question to HE 0435−1223, selected for its resolved four-image geometry and public long-term COSMOGRAIL monitoring [14,15]. The adopted redshifts are $z_l = 0.4546$ and $z_s = 1.693$. HST point-image and galaxy astrometry comes from Courbin et al. [16], with the required additional 2 mas systematic uncertainty. The unavailable full positional covariance is approximated by a diagonal covariance for the published relative positions; we do not fit Einstein-ring pixels or published cosmological posteriors.

The endpoint of the earlier 884-epoch release fixes the chronological boundary at native day 57447.03918. The 2020 release reprocesses some historical photometry, so we use its single consistent reduction: 902 training epochs and 100 later epochs, each containing four R-band magnitudes and their errors. The historical release is not concatenated with the newer reduction. The catalogue and release page differ by half a day in their stated absolute epoch origin; native day differences suffice for all splits and delays here. This is retrospective chronological validation in a common reduction, not a prospectively blinded observing campaign.

The protocol was fixed before fitting. Variability, microlensing and macro parameters were fitted on historical observations only, and the prediction artifact was committed before target scoring. The primary test conditions on all 100 later image-B observations as an explicitly allowed driver and predicts the 300 reserved A/C/D magnitudes. It is a batch cross-image conditional prediction; some driver observations are later than a target date. A separate all-channel forecast reserves all 400 later magnitudes and conditions on training data only. Neither test refits parameters, photometric offsets or extrinsic curves against its targets.

8.2. The conditional physical and brightness applications

We retain the same fixed R1 reference calibration, recomputing its FLRW distance maps and local scalar coefficients at the observed redshifts. This common geometry defines the conventional galaxy-potential benchmark and the conditional R1 comparison; it is not a cosmological viability fit. The local body is a positive Plummer source with fitted mass and scale. A rear quadratic optical screen at the fixed conditional redshift $z_p = 0.8$ supplies asymmetric propagation. It is total optical boundary information, rather than an identified physical perturber or a solved same-domain scalar environment.

For physical screen potential $\widehat{\psi}_p(x_p) = x_p^T U x_p/2$ with symmetric U, the original two-plane ray equations are

$$x_p = D_p\theta - D_{lp}\widehat{\alpha}_l(D_l\theta),$$
$$x_s = D_s\theta - D_{ls}\widehat{\alpha}_l(D_l\theta) - D_{ps}Ux_p. \tag{40}$$

Their elimination gives the compatible maps

$$\mathsf{D}_{ol} = D_l I, \qquad \mathsf{D}_{os} = D_s I - D_{ps}D_p U, \qquad \mathsf{D}_{ls} = D_{ls} I - D_{ps}D_{lp}U. \tag{41}$$

The same eliminated two-plane arrival action gives the time-delay matrix and observer-clock normalization used in Equation (24). No second sheet is added to the local body. Supplementary Note S9 gives the full action and Schur reduction, including the required pre-conjugate map conditions and the isolated spherical scalar boundary prescription.

The local field is still the original regular sourced solution. At the refitted R1 image-A impact $r = 7.74161173$ kpc, the executed section has $\pi_T'/(M_{\mathrm{Pl}}H_0) = -1.83519726 \times 10^{-4}$ and consumed source $J_{\mathrm{eff}}/(M_{\mathrm{Pl}}H_0^2) = -40596.7951$. In $H_0 = M_{\mathrm{Pl}} = 1$ units the integrated scalar left and source rows both give $-2.36867868266 \times 10^{-13}$; the corresponding metric Gauss rows both give $5.16669407301 \times 10^{-12}$. The maximum scalar-flux residual is 1.10×10^{-15} and the four metric-Gauss section residuals are below 3.80×10^{-16} relative. These numerical field residuals do not establish that the assumed galaxy profile is observationally adequate.

For observed magnitude m_{ik} at observer time t_{ik}, we apply the classical variability likelihood

$$m_{ik} = m_0 + b_i + q(t_{ik} - d_i) + e_i(t_{ik}) + \epsilon_{ik}. \tag{42}$$

Here q is a common stationary Ornstein–Uhlenbeck process, a standard empirical quasar-variability model [17]; e_i are independent slower processes for extrinsic variability, and b_i have proper broad Gaussian offset priors. Their covariances are $\sigma_q^2 e^{-|\Delta t|/\tau_q}$ and $\sigma_e^2 e^{-|\Delta t|/\tau_e}$, respectively. Quoted photometric variance plus fitted common jitter supplies the independent measurement noise approximation; the total brightness covariance remains correlated. This empirical likelihood does not replace a BHSM quantum-state kernel.

An exact nine-state Kalman filter evaluates the normalized likelihood. At source time $u = t - d_i$, its state is $(q, e_A, e_B, e_C, e_D, b_A, b_B, b_C, b_D)$, with OU transitions for the first five components and constant offsets. Smoothing over each permitted conditioning set gives predictive means and marginal variances. The joint test score is

$$\log p(y_{\mathrm{target}} \mid y_{\mathrm{condition}}, \widehat{\vartheta}) = \log p(y_{\mathrm{condition}}, y_{\mathrm{target}} \mid \widehat{\vartheta}) - \log p(y_{\mathrm{condition}} \mid \widehat{\vartheta}), \tag{43}$$

with all parameters $\widehat{\vartheta}$ frozen. It is not the sum of univariate marginal scores. Supplementary Note S10 derives these applications and compares them with independent dense-Gaussian formulas.

The empirical training delays relative to A are

$$(d_B, d_C, d_D) = (6.41984, -0.97473, 12.48799) \text{ days},$$

where positive means later arrival. Four fixed starts retain multiple local modes; two reach nearly the same best likelihood. Both fitted variability timescales hit their preset upper bounds, 1500 and 10000 days. Delay-curvature estimates depend materially on step size and are not calibrated confidence limits. The macro fits use the exact historical brightness likelihood after initialization. Their shared variability hyperparameters remain fixed: this

is a two-stage conditional empirical Bayes calculation, rather than a fully marginalized physical and stochastic posterior.

8.3. Frozen predictions and observed performance

The conventional macro fit predicts delays $(6.020014, 0.672696, 13.611423)$ days; the separately refitted R1 model predicts $(6.022659, 0.672780, 13.616498)$ days. Their astrometric residual sums are 12.714 and 12.695 over eight relative-coordinate rows. The conventional fit has $M = 10^{11.61394} M_\odot$, Plummer scale $a = 10^{0.70626}$ kpc and screen convergence 0.3844 within the declared bounds. This large conditional propagation nuisance is not an independently measured line-of-sight convergence.

Activating R1 at exactly the conventional fitted body, source and screen changes the three delays by $(19.2903, 0.9386, 29.0367)$ s. Separately refitted differences, approximately $(228.51, 7.23, 438.44)$ s, also contain nuisance reallocation and optimizer termination effects; they are not the fixed-body physical signal. The independent full two-plane and compact actions agree to 2.4×10^{-14} day. A positive conventional effective mass still reproduces the entire retained R1 transfer, and therefore its test score. That shared score is a structural equality, not an independent empirical confirmation.

Table 7. Frozen later-B-conditioned predictions of 300 withheld A/C/D magnitudes. Log scores are joint continuous predictive densities in the same magnitude units. The zero-delay row keeps the same frozen variability parameters; it is a control, not a separately optimized alternative model. Intervals condition on point-fitted parameters.

Transfer	Joint log score	RMSE (mag)	95% coverage
Free training delays	795.1985	0.101520	85.00%
Conventional macro	791.6238	0.101333	85.33%
Conditional R1 macro	791.6288	0.101333	85.33%
Zero-delay control	755.9755	0.101436	86.67%

The conventional/R1 log-score difference is only 0.00498, while the conventional-minus-control difference is 35.6483 under this frozen likelihood. These are conditional predictive comparisons, not Bayes factors or evidence for a uniquely identified BHSM state. The empirical free-delay baseline scores 3.5746 higher than the restricted conventional macro model. Figure 2 shows that the conventional and R1 curves overlap at plotted precision.

Observed interval coverage is below nominal: conventional image-A coverage is 62%, versus 99% for C and 95% for D. The shared free-delay baseline also undercovers A. This is a limitation of the brightness/uncertainty baseline, rather than a BHSM-specific residual. The all-channel forecast has RMSE 0.194945 mag and 75.5% coverage; the R1-minus-conventional log-score difference is again only 0.00501. These two factorizations use the same later observations and are not independent replications. Temporally correlated targets do not justify an independent-binomial significance calculation for these coverage fractions. The model is not retuned after this exposure; a revised variability model needs fresh reserved data for an independent test.

A broad stationary-image search, with original-projection refinement, finds five images in both models, including a central image with macro flux approximately 6.32% of A before microlensing, extinction and galaxy-light contamination. No measured central-image limit enters the point-astrometry fit. Consequently this application is a restricted timing/point-geometry baseline, not a complete observed galaxy or Einstein-ring reconstruction. A full imaging likelihood must test that additional prediction. Neither its existence nor the held-out brightness residuals are silently absorbed into a new envelope response.

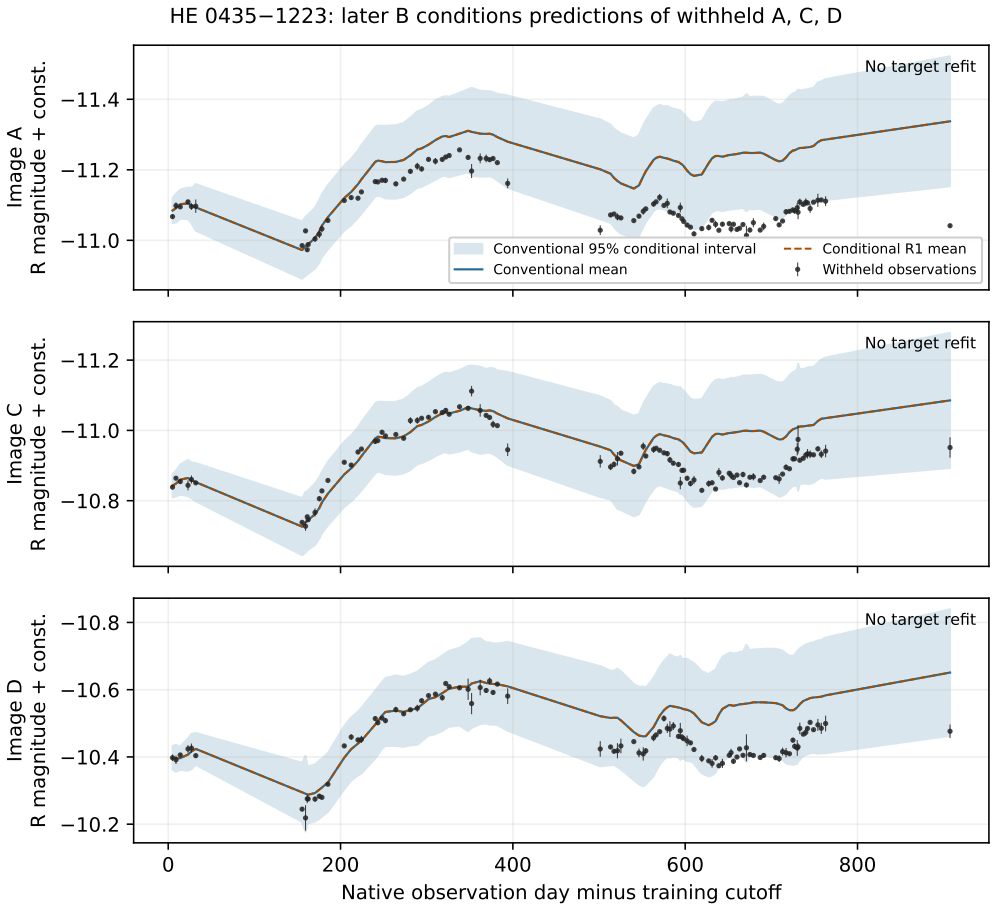


Figure 2. Reserved HE 0435−1223 A/C/D magnitudes and predictions conditioned on historical data and the allowed later-B driver. Lines join frozen predictive means at observation times; shading is the conventional 95% point-parameter conditional interval. No target fit or per-image magnitude correction is applied. Photometry: COSMOGRAIL/CDS [15].

9. Computational Methods and Reproducibility

The calculation uses the fixed manuscript source revision `49a1dab` and records full input hashes. The main reproduction command from the repository root is

```
python scripts/run_strong_lensing_r1_screened.py
```

The coefficient, independent projection, and multiple-source checks are under `scripts/r1_screened_review/`. Numerical results and their units are under `artifacts/strong_lensing_r1_screened_20261020/`. The research package additions and exact commands are integrated with the pinned repository background and optical dependencies. The recorded package-era replay of three portable review scripts reproduced all 305 saved numerical values in that environment. The current integration replayed nine numerical commands, compared 1161 saved numerical values, and passed 16 focused checks without changing the original numerical artifacts. These are separate execution records, not independent physical measurements.

The additional inverse and profile applications are reproduced with `scripts/run_strong_lensing_refit.py` and `scripts/run_strong_lensing_robustness.py`; the distributed optical control uses `scripts/run_strong_lensing_encounter_accuracy.py`. Their outputs are under `artifacts/strong_lensing_acceptance_20261020/`. Four refit, seven profile/coefficient and seven encounter tests check the original applications and independent projections. The distributed-control receipt records its boundary identities, original variational residuals and convergence checks. The complete commands and software versions are in the acceptance checkpoint. The original 305-value portable package check and the current 1161-value replay remain separate records.

The reproducibility manifest identifies versions and hashes rather than equating successful execution with physical validation.

The independent quadrature uses the same physical radial equation but a separately implemented integration variable and root calculation. Its agreement is a reproducibility check, not independent empirical evidence or external human peer review.

The observed application archives original COSMOGRAIL/CDS files and row-level provenance, the pre-fit protocol, all four training starts, the committed pre-scoring predictions and the single reserved-target application. Eight light-curve and seven macro tests check exact likelihood normalization, dense conditioning, target-label invariance, delay sign/recovery, compatible maps, two-plane action reduction and consistent radial projection. The executed application and forward-audit receipts verify original scalar-source and metric-Gauss residuals. The full source and conditional-statistics definitions are in Supplementary Notes S9–S10. The scoring-start record persists even after an aborted exposure, so these targets cannot silently be reclassified as unseen after model revision. A separately labeled known-outcome replay reproduces the saved predictions and scores without refitting or reopening the experiment; it is not a second independent test.

10. Discussion: State Inference and Physical Scope

The calculation answers one forward and inverse modeling question. A specified R1 state and finite source determine a regular encounter and its light observables. The coefficient derivatives quantify that state dependence. A positive effective mass represents the same retained metric, and a finite conventional nuisance fit reproduces the injected three-image system. Accordingly, the reference-relative seconds are a calculated response at fixed preparation; assigning them to a unique envelope state requires restrictions on the ordinary lens model and the state-to-source relation. A shared multiple-source profile can retain information that a one-source fit absorbs, but the complete positive effective-mass representation remains available at this order. This is a constructive equivalence within the BHSM transfer program.

The smooth-profile comparison gives the practical scale of this issue. The imposed profile changes alter ordinary delays much more than the state-conditioned offsets. They do not define an empirical prior; they show why a model-conditioned transfer is insufficient on its own for state identification. The finite refit is likewise a saturated local example rather than a forecast. The HE 0435 application supplies an actual but restricted test using point imaging and original brightness likelihoods. Its additional central-image prediction and conditional optical screen prevent interpreting that point fit as a complete measured galaxy or independently reconstructed environment.

The executed chronological test illustrates how existing light can establish a per-lens baseline, and how reserved observations can test it. Its conventional/R1 predictive tie is consistent with the effective-mass equivalence. Its brightness undercoverage, particularly in image A, shows why recovering a transfer does not establish a calibrated microlensing/source-variability model or unique physical state. The two-stage point-fit intervals exclude parameter uncertainty; the observed coverage is descriptive for correlated observations. No later residual is assigned to envelope evolution or a new interface amplitude.

A progressively constrained optical map needs the admitted mass, source, line-of-sight and calibration uncertainties and a specified interface law for an interface-state inference. A published cosmological posterior conditioned on a lens model cannot replace the light-data likelihood. For HE 0435, a revised variability law or spatial profile would need new reserved observations for an independent predictive test. The positive conventional representation must remain available in that inference. This is the same transfer-to-identifiability question tested first in the controlled finite-body calculation.

The approximation controls have separate meanings. The common GR2 term moves absolute image coordinates by up to 1.72 microarcsec, while its change of the reference-relative image offsets is below 7×10^{-6} microarcsec and its delay-offset change is below 0.1 ms. The distributed-encounter control tests the optical plane within the retained leading action. Its finite-window and path effects are distinct from scalar-dependent field-equation higher orders. The local parameter $H_l r_V/c = 5.64 \times 10^{-5}$ and the 1.1–1.7 $\times 10^{-6}$ fraction of extra deflection beyond 20 Mpc in the formal isolated projection characterize scale separation and tails. They are not complete physical error bounds. A time-dependent material/environmental continuation consumes the specific equation of state or support and constrained outer data derived in Supplementary Note S5.

The leading scalar principal coefficients are positive on the retained branch (Appendix C); its maximum radial characteristic speed squared is approximately 1.474 in metric light-speed units. Photons in this benchmark follow metric null rays. This leading coefficient check establishes neither full coupled stability nor a causal ultraviolet completion. The independent rolling scalar T, target phase and metric coordinate retain their stated normalizations; the earlier source-free target null is preserved in Appendix A.

The BHSM microscopic source question has a precise boundary in this paper. Supplementary Note S4 derives the cap/matcher and Schur source pullback and gives positive same-domain complements with identical leading normalization but different higher-derivative coefficients. A physical cap/interface embedding into canonical T, its cubic vertex, source pairing and complement length/spectrum must therefore be supplied by the framework before a unique microscopic state is assigned to this transfer. This result supplies the defining conditions and their nonuniqueness; it does not select an extra optical amplitude. The present calculation is a reproducible effective benchmark against which that physical source construction can be tested.

11. Conclusions

For the prescribed $10^{11} M_\odot$, 3 kpc body at $z_l = 0.3$ and common source at $z_s = 1$, the fixed R1 action/state produces a regular source, screening radius approximately 212 kpc, reference-relative delay offsets 25.96 s and 8.65 s, image changes approximately 0.79–6.12 microarcsec and a 6.82 ppm delay-ratio change. The state dependence enters explicitly through the normalized scalar mixing and stiffness. These results retain one physical incident-light, clock and propagation prescription throughout.

The same retained metric and light transfer admit a positive conventional effective mass. A finite five-parameter ordinary refit also reproduces the three injected images and two delays to numerical precision; fixing its center leaves millisecond-scale delay mismatches in the declared profile family. The interpretation of the response therefore depends on the admitted lens freedoms. A smooth Plummer control matched by total mass and half-mass radius gives offsets 22.45 s and 9.62 s while preserving the effective-mass equivalence. The eight paired profile/input controls, common ordinary-GR correction and distributed optical encounter make the benchmark's model and approximation dependencies explicit. The common GR2 changes of the original offsets remain $+69$ and -10 microseconds.

The observed HE 0435 baseline trains on 902 epochs and reserves 100 later epochs. With later B as the allowed driver, conventional and R1 predictions for 300 A/C/D targets differ by only 0.00498 in joint log score. Their nominal 95% intervals cover 85.33% of these targets, and the separately retained all-channel forecast also undercovers. The experiment supplies a reproducible observed timing/point-geometry test and exposes its brightness and central-image limitations. It does not identify a unique BHSM envelope state.

Prepared light constrains an envelope encounter transfer. Identifying that transfer with a particular underlying state consumes the specified action/source map and independently justified mass/environment data. The calculation provides a coherent forward and inverse test problem for that inference, with its physical source conditions and technical continuation derived separately.

Supplementary Materials: The repository supplies Notes S1–S10: local action/source reduction; finite-body GR2; nuisance optics; cap/interface source pullback; exact field/environmental continuation; profile robustness; finite nonlinear refit; distributed-encounter accuracy control; observed-lens source/two-plane derivation; and classical light-curve likelihood and reserved-target protocol. Their source paths, executable commands, input hashes, numerical outputs and focused checks are mapped in the acceptance checkpoint. The long cap and continuation TeX derivations are retained as technical supplementary source; the historical target and transport calculations remain available.

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COSMOGRAIL/CDS release cited above; original files, attribution, hashes and reduction caveats are retained in the data provenance.

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Appendix A. The Constant-Target Null Control

The earlier source-free target sector has

$$S_\eta = -\int N\sqrt{h}\, w_0 F(X)\, \mathrm{d}t\, \mathrm{d}^3x, \qquad F(X) = \frac{\kappa_\eta}{2}X + \frac{1}{8}X^4, \qquad X = g^{\mu\nu}\langle D_\mu\eta, D_\nu\eta\rangle, \quad \text{(A1)}$$

with $\eta = (s,z) \in S^6 \subset \mathbb{R}\oplus\mathbb{C}^3$, $s^2 + 2z^\dagger z = 1$, target metric $\mathrm{d}s^2 + 2\,\mathrm{Re}(\mathrm{d}z^\dagger \mathrm{d}z)$, and $w_0\kappa_\eta > 0$. For $\eta_q = (\cos q, \sin q\, e_1/\sqrt{2})$ with constant q and zero connection, $D\eta = 0$, $F(0) = 0$, and the target current and stress vanish. For a radial meridian $q(r)$, regularity sets the conserved flux

$$r^2 e^\nu \sqrt{f_r}\, w_0(\kappa_\eta + X^3)q' = 0, \qquad X = f_r(q')^2 \quad \text{(A2)}$$

to zero. Positivity gives $q' = 0$. This proves the null only for the stated source-free regular boundary problem.

A constant background still permits positive perturbation stiffness. For a nonconstant Dirichlet phase perturbation, the quadratic energy and its second variation (phase self-curvature) are

$$E_\eta^{(2)}[\varphi] = \tfrac{1}{2} w_0\kappa_\eta \int e^{\Phi-\Psi}|\nabla\varphi|^2\, \mathrm{d}^3x,$$

$$\delta^2 E_\eta[\varphi,\varphi] = 2E_\eta^{(2)}[\varphi] > 0.$$

The mixed metric/phase response vanishes at $D\eta = 0$. The solved matter body therefore has ordinary nonzero lensing and zero additional response in this target family. A vanishing matched seam contribution is not a statement that all possible gravitational or boundary sources vanish. The rolling scalar T in Equation (4) is a different explicitly specified field.

Appendix B. Moving Measured-Redshift Endpoints

For an observer-normalized affine ray, $J_a'' = \mathcal{T} J_a$ with $(J_a, J_a')_a = (0, \omega_a I)$. Let $Z_{\lambda,a} = \delta \ln\omega|_{\lambda_a}$ and $Z_{\chi,a} = \delta\ln\omega|_{\chi_a}$. On the reference ray,

$$Z_{\chi,a} = Z_{\lambda,a} - \mathcal{H}_a \delta\chi_{\mathrm{ray},a}|_\lambda,$$

$$\delta\lambda_a = -\frac{Z_{\lambda,a}}{(\ln\overline{\omega})'_{\lambda,a}}, \qquad \delta\chi_a = \delta\chi_{\mathrm{ray},a}|_\lambda + \overline{\chi}'_{\lambda,a}\delta\lambda_a = -\frac{Z_{\chi,a}}{\mathcal{H}_a}. \quad \text{(A3)}$$

Thus converting an affine shift alone does not give the full radial event shift. The moving-vertex initial conditions are

$$\delta J_a|_a = -\omega_a I\, \delta\lambda_a, \qquad \delta J_a'|_a = \Delta\omega_a I, \qquad \Delta\omega_a = \delta\omega_a|_\lambda + \overline{\omega}'_a\delta\lambda_a. \quad \text{(A4)}$$

At fixed measured redshift and observer normalization, $\Delta\omega_a = 0$. Output event and frame conversions must also be included. These conventions are needed for the three compatible maps; they are inactive in the local comparison where the maps are held fixed.

Appendix C. Leading Scalar Principal Coefficients

After eliminating the leading metric mixing, the homogeneous time coefficient is

$$D = 1 + \frac{12Hv}{\lambda} + \frac{6v^4}{M_{\rm Pl}^2\lambda^2} = 4.3962210287. \tag{A5}$$

For $\mathcal{S} = \sqrt{C^2 + 16b_T GM(<r)/(\lambda r^3)}$, the leading local diagonal coefficients are

$$K_r = \mathcal{S}, \qquad K_\Omega = \mathcal{S} + \frac{1}{2}r\mathcal{S}', \qquad K_t = D + \frac{3}{2}(\mathcal{S} - C) + \frac{1}{2}r\mathcal{S}'. \tag{A6}$$

Nonnegative density implies $K_\Omega \geq (\mathcal{S}^2 + 3C^2)/(4\mathcal{S}) > 0$. For this branch K_t is also positive; the frozen homogeneous ratio is $C/D = 0.9642051$. The exterior maximum of K_r/K_t is 1.47366 at approximately 100.107 kpc. These are leading local coefficients. Mixed time–space and higher weak-field terms, the full coupled spectrum, and an ultraviolet completion are outside this check.

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